

Differentiate Automatically

Automatic Differentiation (Part I)

Jonathon Hare

Vision, Learning and Control
University of Southampton

Much of this material is based on this blog post:

<https://rufflewind.com/2016-12-30/reverse-mode-automatic-differentiation>

What is Automatic Differentiation (AD)?

To solve optimisation problems using gradient methods we need to compute the gradients (derivatives) of the objective with respect to the parameters.

- In neural nets we're talking about the gradients of the loss function, $\mathcal{L}$ with respect to the parameters θ : $\nabla_{\theta}\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\theta}$
- AD is important - it's been suggested that "Differentiable programming" could be the term that ultimately replaces deep learning¹.

¹<http://forums.fast.ai/t/differentiable-programming-is-this-why-we-switched-to-pytorch/9589/5>

How can you compute derivatives?

There are three ways to compute derivatives:

- Symbolically differentiate the function with respect to its parameters
 - by hand
 - using a CAS
- Make estimates using finite differences
- Use Automatic Differentiation

How can you compute derivatives?

There are three ways to compute derivatives:

- Symbolically differentiate the function with respect to its parameters
 - by hand
 - using a CAS
- Make estimates using finite differences
- Use Automatic Differentiation

Problems

Static - can't “differentiate algorithms”

How can you compute derivatives?

There are three ways to compute derivatives:

- Symbolically differentiate the function with respect to its parameters
 - by hand
 - using a CAS
- Make estimates using finite differences
- Use Automatic Differentiation

Problems

Numerical errors - could compound in deep nets; expensive

How can you compute derivatives?

There are three ways to compute derivatives:

- Symbolically differentiate the function with respect to its parameters
 - by hand
 - using a CAS
- Make estimates using finite differences
- Use Automatic Differentiation

Numerical Differentiation

What is Automatic Differentiation (AD)?

Automatic Differentiation is:

- a method to get exact derivatives efficiently, by storing information as you go forward that you can reuse as you go backwards.
 - Takes code that computes a function and uses that to compute the derivative of that function.
 - The goal isn't to obtain closed-form solutions, but to be able to write a program that efficiently computes the derivatives.

Lets think about differentiation and programming

Example (Math)

$$x = ?$$

$$y = ?$$

$$a = x y$$

$$b = \sin(x)$$

$$z = a + b$$

Lets think about differentiation and programming

Example (Math)

$x = ?$

$y = ?$

$a = x y$

$b = \sin(x)$

$z = a + b$

Example (Code)

```
x = ?
```

```
y = ?
```

```
a = x * y
```

```
b = sin(x)
```

```
z = a + b
```

The Chain Rule of Differentiation

Recall the chain rule for a variable/function z that depends on y which depends on x :

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$$

The Chain Rule of Differentiation

Recall the chain rule for a variable/function z that depends on y which depends on x :

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$$

In general, the chain rule can be expressed as:

$$\frac{\partial w}{\partial t} = \sum_i^N \frac{\partial w}{\partial u_i} \frac{\partial u_i}{\partial t} = \frac{\partial w}{\partial u_1} \frac{\partial u_1}{\partial t} + \frac{\partial w}{\partial u_2} \frac{\partial u_2}{\partial t} + \cdots + \frac{\partial w}{\partial u_N} \frac{\partial u_N}{\partial t}$$

where w is some output variable, and u_i denotes each input variable w depends on.

Applying the Chain Rule

Let's differentiate our previous expression with respect to some yet to be given variable t :

Expression

$$x = ?$$

$$y = ?$$

$$a = x y$$

$$b = \sin(x)$$

$$z = a + b$$

$$\frac{\partial x}{\partial t} = ?$$

$$\frac{\partial y}{\partial t} = ?$$

$$\frac{\partial a}{\partial t} = x \frac{\partial y}{\partial t} + y \frac{\partial x}{\partial t}$$

$$\frac{\partial b}{\partial t} = \cos(x) \frac{\partial x}{\partial t}$$

$$\frac{\partial z}{\partial t} = \frac{\partial a}{\partial t} + \frac{\partial b}{\partial t}$$

Applying the Chain Rule

Let's differentiate our previous expression with respect to some yet to be given variable t :

Expression

$$x = ?$$

$$y = ?$$

$$a = x y$$

$$b = \sin(x)$$

$$z = a + b$$

$$\frac{\partial x}{\partial t} = ?$$

$$\frac{\partial y}{\partial t} = ?$$

$$\frac{\partial a}{\partial t} = x \frac{\partial y}{\partial t} + y \frac{\partial x}{\partial t}$$

$$\frac{\partial b}{\partial t} = \cos(x) \frac{\partial x}{\partial t}$$

$$\frac{\partial z}{\partial t} = \frac{\partial a}{\partial t} + \frac{\partial b}{\partial t}$$

If we substitute $t = x$ in the above we'll have an algorithm for computing $\partial z / \partial x$. To get $\partial z / \partial y$ we'd just substitute $t = y$.

Translating to code I

We could translate the previous expressions back into a program involving *differential variables* $\{dx, dy, \dots\}$ which represent $\partial x / \partial t, \partial y / \partial t, \dots$ respectively:

```
dx = ?  
dy = ?  
da = y * dx + x * dy  
db = cos(x) * dx  
dz = da + db
```


Translating to code I

We could translate the previous expressions back into a program involving *differential variables* $\{dx, dy, \dots\}$ which represent $\partial x / \partial t, \partial y / \partial t, \dots$ respectively:

```
dx = ?  
dy = ?  
da = y * dx + x * dy  
db = cos(x) * dx  
dz = da + db
```

What happens to this program if we substitute $t = x$ into the math expression?

Translating to code II

```
dx = 1
dy = 0
da = y * dx + x * dy
db = cos(x) * dx
dz = da + db
```

Translating to code II

```
dx = 1
dy = 0
da = y * dx + x * dy
db = cos(x) * dx
dz = da + db
```

The effect is remarkably simple:
to compute $\partial z / \partial x$ we just seed
the algorithm with $dx=1$ and
 $dy=0$.

Translating to code III

```
dx = 0
dy = 1
da = y * dx + x * dy
db = cos(x) * dx
dz = da + db
```

To compute $\partial z / \partial y$ we just seed the algorithm with $dx=0$ and $dy=1$.

Making Rules

- We've successfully computed the gradients for a specific function, but the process was far from automatic.

Making Rules

- We've successfully computed the gradients for a specific function, but the process was far from automatic.
- We need to formalise a set of rules for translating a program that evaluates an expression into a program that evaluates its derivatives.

Making Rules

- We've successfully computed the gradients for a specific function, but the process was far from automatic.
- We need to formalise a set of rules for translating a program that evaluates an expression into a program that evaluates its derivatives.
- We have actually already discovered 3 of these rules:

$$c = a + b \quad \Rightarrow \quad dc = da + db$$

$$c = a * b \quad \Rightarrow \quad dc = b * da + a * db$$

$$c = \sin(a) \quad \Rightarrow \quad dc = \cos(a) * da$$

More rules

These initial rules:

$$c = a + b \quad \Rightarrow \quad dc = da + db$$

$$c = a * b \quad \Rightarrow \quad dc = b * da + a * db$$

$$c = \sin(a) \quad \Rightarrow \quad dc = \cos(a) * da$$

can easily be extended further using multivariable calculus:

$$c = a - b \quad \Rightarrow \quad dc = da - db$$

$$c = a / b \quad \Rightarrow \quad dc = da / b - a * db / b ** 2$$

$$c = a ** b \quad \Rightarrow \quad dc = b * a ** (b - 1) * da + \log(a) * a ** b * db$$

$$c = \cos(a) \quad \Rightarrow \quad dc = -\sin(a) * da$$

$$c = \tan(a) \quad \Rightarrow \quad dc = da / \cos(a) ** 2$$

- To translate using the rules we simply replace each primitive operation in the original program by its differential analogue.

- To translate using the rules we simply replace each primitive operation in the original program by its differential analogue.
- The order of computation remains unchanged: if a statement K is evaluated before another statement L , then the differential analogue of K is evaluated before the analogue statement of L .

- To translate using the rules we simply replace each primitive operation in the original program by its differential analogue.
- The order of computation remains unchanged: if a statement K is evaluated before another statement L , then the differential analogue of K is evaluated before the analogue statement of L .
- This is **Forward-mode Automatic Differentiation**.

Interleaving differential computation

A careful analysis of our original program and its differential analogue shows that its possible to interleave the differential calculations with the original ones:

$$x = ?$$
$$dx = ?$$
$$y = ?$$
$$dy = ?$$
$$a = x * y$$
$$da = y * dx + x * dy$$
$$b = \sin(x)$$
$$db = \cos(x) * dx$$
$$z = a + b$$
$$dz = da + db$$

Interleaving differential computation

A careful analysis of our original program and its differential analogue shows that its possible to interleave the differential calculations with the original ones:

`x = ?`

`dx = ?`

`y = ?`

`dy = ?`

`a = x * y`

`da = y * dx + x * dy`

`b = sin(x)`

`db = cos(x) * dx`

`z = a + b`

`dz = da + db`

Dual Numbers

- This implies that we can keep track of the value and gradient at the same time.

Interleaving differential computation

A careful analysis of our original program and its differential analogue shows that its possible to interleave the differential calculations with the original ones:

$x = ?$

$dx = ?$

$y = ?$

$dy = ?$

$a = x * y$

$da = y * dx + x * dy$

$b = \sin(x)$

$db = \cos(x) * dx$

$z = a + b$

$dz = da + db$

Dual Numbers

- This implies that we can keep track of the value and gradient at the same time.
- We can use a mathematical concept called a “Dual Number” to create a very simple direct implementation of AD.